ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 3

INTRODUCTION TO TECHNICAL PROGRAMMING

PAPER 1 · PRACTICE D

MARKS: 120 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 3. Laid out like the DHET first paper (120 marks). Section A (31): multiple choice, true or false, matching C++ terms and labelling the parts of a program · Section B (89): programming concepts and computational thinking, Polya's problem-solving steps, an mBlock times-table program to study (with C++ and a flowchart) and a C++ tuck-shop program to study.

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 3. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. Program code in this paper is written in C++, except for the mBlock program in QUESTION 7. Code is not marked strictly on syntax.
6. Write neatly and legibly.

SECTION A

QUESTION 1

Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1–1.10) in the ANSWER BOOK.

1.1 Which one of the following is a C++ integrated development environment (IDE)?

A Excel
B PowerPoint
C Paint
D Code::Blocks

1.2 The part of a computer that holds a program and its data while the program is running is the …

A hard drive.
B RAM.
C monitor.
D keyboard.

1.3 Which C++ operator tests whether two values are equal?

A ==
B =
C !=
D :=

1.4 What does the statement cout << 7 / 2; display when 7 and 2 are integers?

A 3.5
B 4
C 3
D 2

1.5 Which Git command saves staged changes as a new version in the local repository?

A git clone
B git commit
C git pull
D git init

1.6 A step-by-step plan written in everyday language, set out like a program, is called …

A pseudocode.
B machine code.
C binary.
D HTML.

1.7 Which C++ data type should be used to store the price R19.99?

A int
B char
C bool
D double

1.8 In mBlock, the "forever" and "repeat" blocks are found in the … category.

A Looks
B Sound
C Control
D Operators

1.9 Which symbols start a comment that can cover several lines in C++?

A /*
B //
C #
D --

1.10 Which loop is the best choice when the number of repetitions is known in advance?

A while
B for
C do … while
D if

(10)

[10]

QUESTION 2

Indicate whether the following statements are TRUE or FALSE. Write only 'True' or 'False' next to the question number (2.1–2.5) in the ANSWER BOOK.

2.1 An interpreter translates and runs a program one statement at a time.

2.2 In C++, the value of 10 % 3 is 3.

2.3 Many programmers can share and work on the same GitHub repository.

2.4 In C++, a string must be written between single quotes.

2.5 Indenting the code inside loops and if statements makes a program easier to read.

(5)

[5]

QUESTION 3

Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (3.1–3.10) in the ANSWER BOOK.

	COLUMN A		COLUMN B
3.1	Data type that holds text such as a name	A	||

3.2	Reads a value typed by the user	B	cin
3.3	Program that turns C++ source code into machine code	C	string
3.4	Logical AND operator	D	=
3.5	Runs a block of code only when a condition is true	E	compiler
3.6	Makes a loop jump to its next repetition	F	continue
3.7	Not equal to	G	&&
3.8	Joins conditions so that only one of them needs to be true	H	0
3.9	Value returned by main() when the program ends successfully	I	if
3.10	Gives a variable a new value	J	!=

(10)

[10]

QUESTION 4

Study the C++ program below. Name the part of the program on each line marked A–F (4.1–4.6).

```
A   #include <iostream>
B   using namespace std;
C   int main() {
D       int age = 18;
E       cout << "Age: " << age << endl;
F       return 0;
    }
```

4.1 A

4.2 B

4.3 C

4.4 D

4.5 E

4.6 F

(6)

[6]

TOTAL SECTION A: [31]

SECTION B

QUESTION 5

5.1 Answer the questions below. Write only the answer next to the question number (5.1.1–5.1.5).

5.1.1 Define the term logic error. (2)

5.1.2 Define the term constant, and give an example of a constant declared in C++. (4)

5.1.3 Name the main circuit board that connects all the parts of a computer. (2)

5.1.4 Name the component that changes the digital signals of a computer into the picture on the screen. (2)

5.1.5 Define the term debugging. (2)

5.2 Define the term iteration. (2)

5.3 Name the FOUR main ideas (cornerstones) of computational thinking. (4)

5.4 Explain the difference between a function that returns a value and a void function. (4)

5.5 Give TWO advantages of block-based programming (such as mBlock) for beginners. (4)

[26]

QUESTION 6

Answer the following question as fully as possible.

> The owner of a college tuck shop wants a program that works out how much change to give a customer, and how many R10 notes, R5 coins and R1 coins to use. You have been asked to develop the program.

Explain how you would use Polya's FOUR problem-solving steps to develop this program. Refer to the tuck-shop problem in each step. (11)

[11]

QUESTION 7

Study the mBlock program below and answer the questions.

```
when green flag clicked
set [number] to 0
set [count] to 1
ask [Which times table do you want to practise (1 to 12)?] and wait
if < < (answer) < 1 > or < (answer) > 12 > > then
    say [Choose a number from 1 to 12] for 2 seconds
    stop [all]
set [number] to (answer)
repeat 10
    say (join (number) (join [ x ] (join (count) (join [ = ] ((number) * (count)))))) for 1 seconds
    change [count] by 1
end repeat
say [Done!] for 2 seconds
```

7.1 Name the TWO variables in the program and say what each one stores. (2)

7.2 How many sums does the sprite say, and which block decides this? (2)

7.3 What is the purpose of the "if … then" block? (2)

7.4 The user enters 7. What is the first sum that the sprite says? (2)

7.5 The user enters 5. What is the last sum that the sprite says before "Done!"? (2)

7.6 Write a C++ program that does the same as the mBlock program: it asks for a times table from 1 to 12, shows a message if the number is out of range, and otherwise uses a for loop to display the ten sums. (8)

7.7 Why does the variable count start at 1 and not at 0? (2)

7.8 Which block must be changed so that the program shows the sums up to × 12, and how? (2)

7.9 Draw a flowchart for the mBlock program above. (10)

[32]

QUESTION 8

> A college tuck shop sells pies (R18), juice (R12) and chips (R15). A cashier types the number of each item and its quantity, one item at a time, and enters 0 to finish. Bills of more than R100 get a 10% discount.

Study the program below and answer the questions that follow.

```
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int choice;
6      int qty;
7      double bill = 0;
```

```
 8
 9      do {
10          cout << "1 Pie R18   2 Juice R12   3 Chips R15   0 Pay" << endl;
11          cout << "Choice: ";
12          cin >> choice;
13          if (choice == 0) break;
14          cout << "Quantity: ";
15          cin >> qty;
16          switch (choice) {
17              case 1: bill += 18 * qty; break;
18              case 2: bill += 12 * qty; break;
19              case 3: bill += 15 * qty; break;
20              default: cout << "Not on the menu" << endl;
21          }
22      } while (true);
23
24      if (bill > 100) {
25          bill = bill * 0.9;            // 10% discount
26      }
27      cout << "Amount to pay: R" << bill << endl;
28      return 0;
29  }
```

8.1 Which loop structure is used, and how does the loop end? (2)

8.2 What does the switch statement in lines 16–21 do? (2)

8.3 What is the purpose of the word break in lines 17–19? (2)

8.4 When will the default part (line 20) run? (2)

8.5 Calculate the amount to pay for 2 pies and 1 juice. (2)

8.6 Calculate the amount to pay for 4 pies and 3 chips. (2)

8.7 What does the operator += do in line 17? (2)

8.8 Why is the check for 0 (line 13) done before the quantity is asked for? (2)

8.9 Give the data type of the variable bill and explain why this type is used. (2)

8.10 When a cashier enters 5, the program still asks for a quantity. Suggest how the program can be improved. (2)

[20]

TOTAL SECTION B: [89]

TOTAL: 120

END OF QUESTION PAPER